Infosys - Coding 15

HackerRank

You are given a binary string S consisting of '0's and '1's. Your task is to find the minimum number of adjacent swaps required to group all identical characters into a single contiguous block. You have two options: 1. Group all '1's into one contiguous block. 2. Group all '0's into one contiguous block. Find the minimum number of swaps between these two options. For example, if S = "1001", you could group the '1's into 0110 (1 swap) or group the '0's into 1001 (0 swaps). The minimum is 0.

Input Format

The first line contains a binary string S.

Constraints

1 <= |S| <= 10^5

Output Format

The output contains an integer value

Sample Input 0

```
10101
```

Sample Output 0

```
1
```

Explanation 0

Grouping the three '1's needs 2 swaps (adjusted positions [0,1,2], median 1 → 1+0+1). Grouping the two '0's needs only 1 swap (e.g., within "101"). The overall minimum is 1.

Sample Input 1

```
1001
```

Sample Output 1

```
0
```

Explanation 1

Grouping the two '1's costs 1 swap, but the two '0's are already contiguous ("00" inside 1001), costing 0 swaps. The overall minimum is 0.

Sample Input 2

```
111111101111111101111011
```

Sample Output 2

```
12
```

Explanation 2

There are 21 ones and 3 zeros. Grouping all 21 ones costs 15 swaps. For the zeros: positions are 7, 16, 21 → adjusted [7, 15, 19], median 15 → |7−15| + |15−15| + |19−15| = 8 + 0 + 4 = 12. The overall minimum is 12.